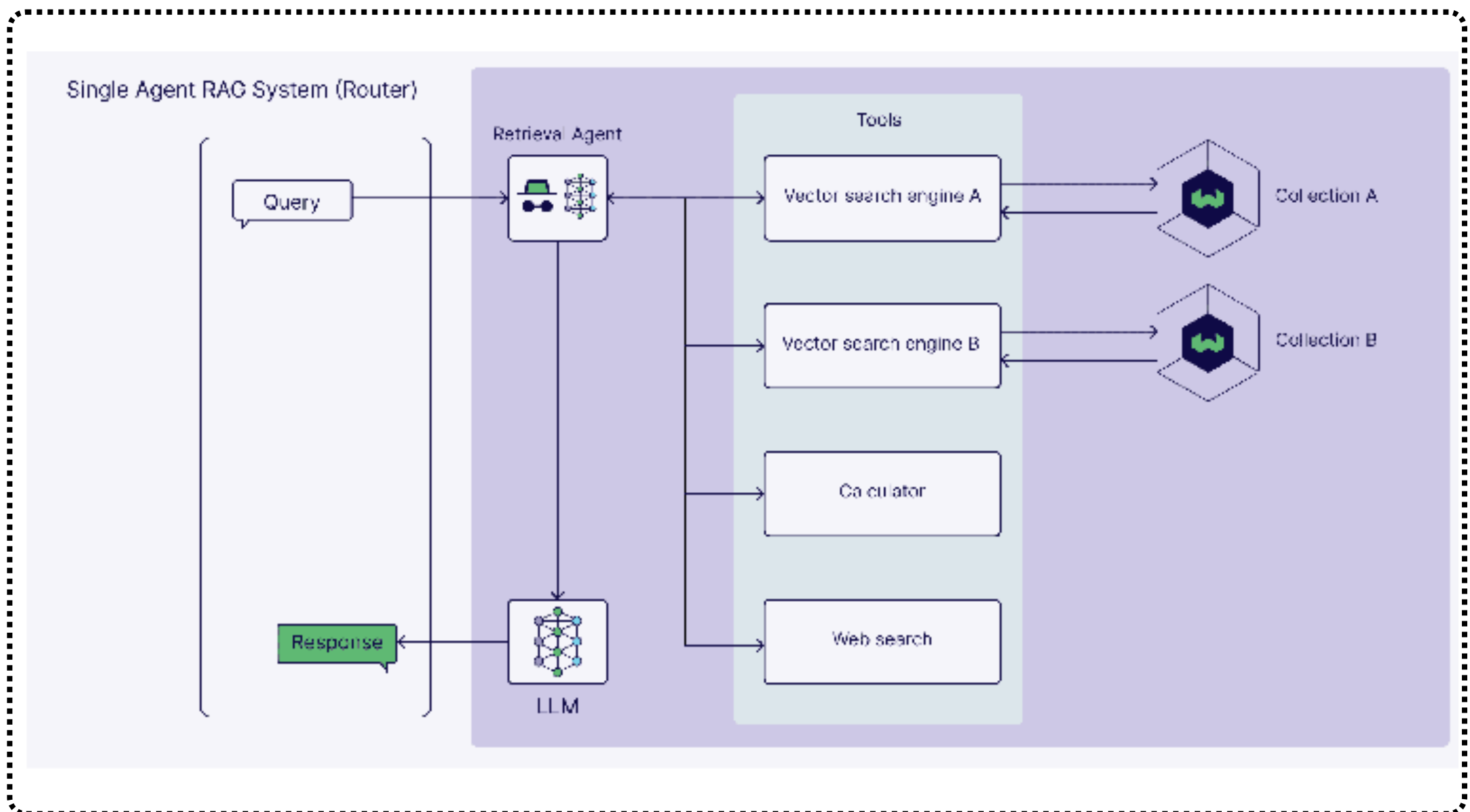
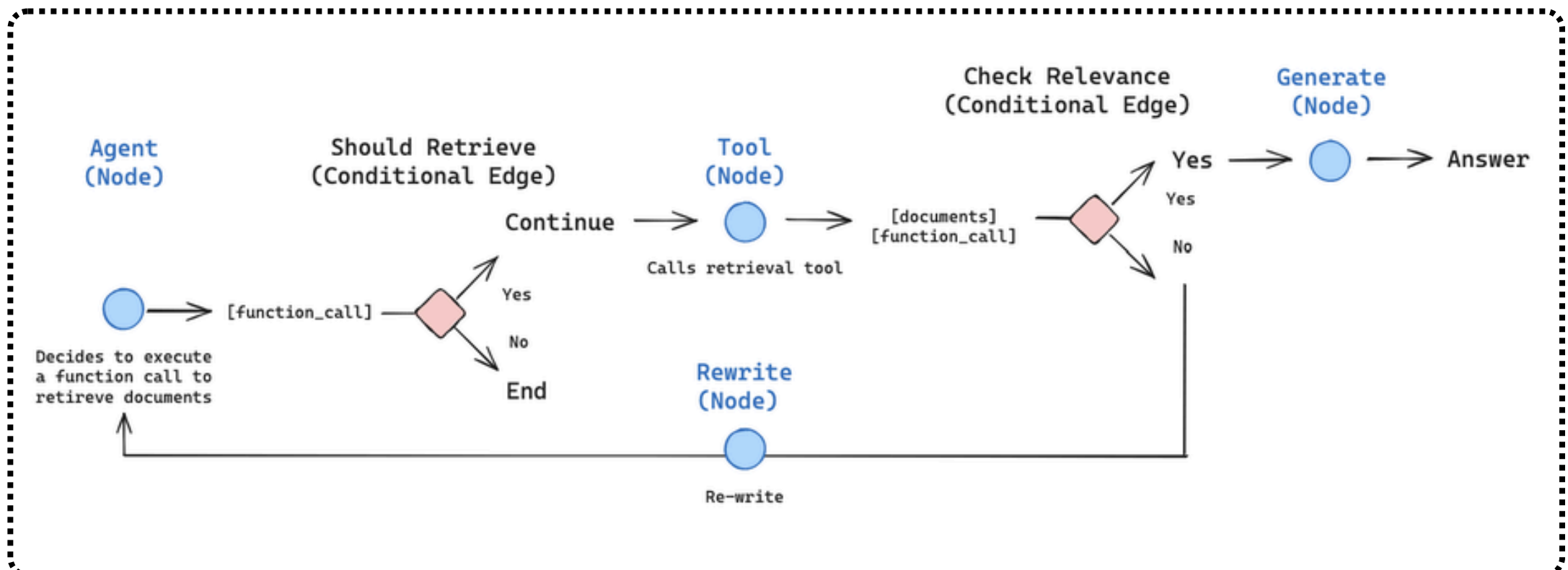


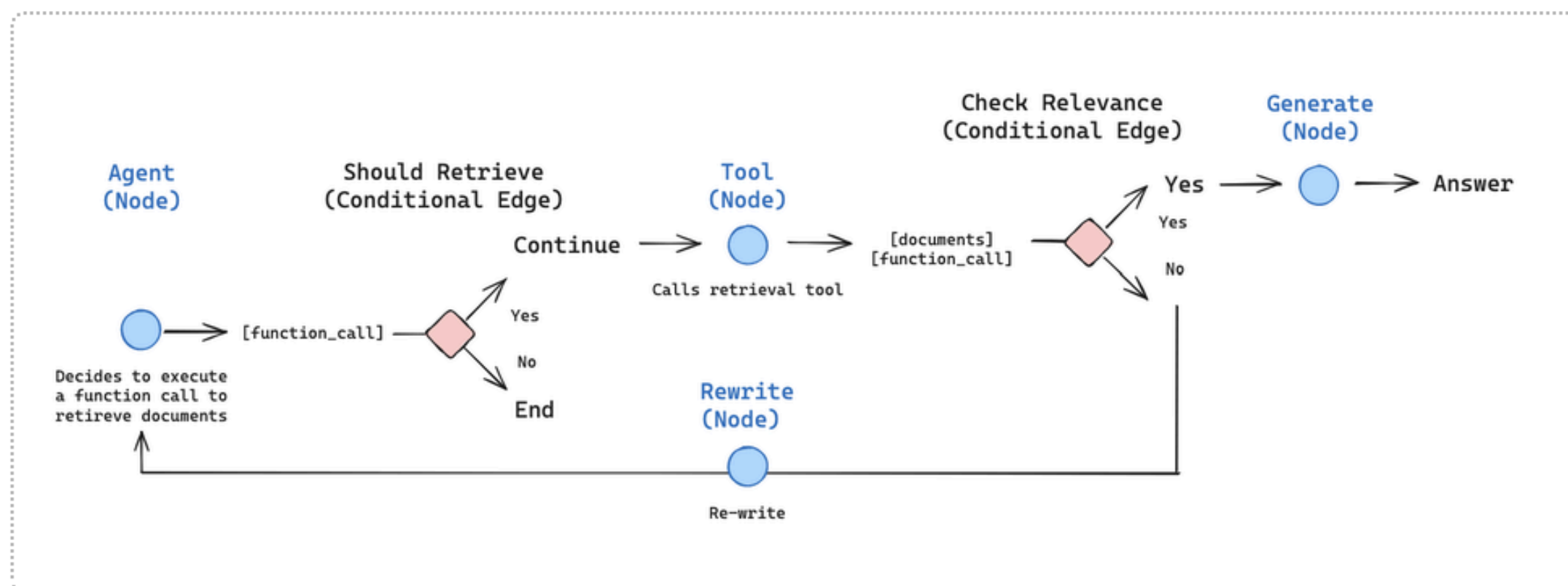
Day 15 of Agentic AI

Agentic RAG



#30DaysofAgenticAI

Unlocking Agentic RAG



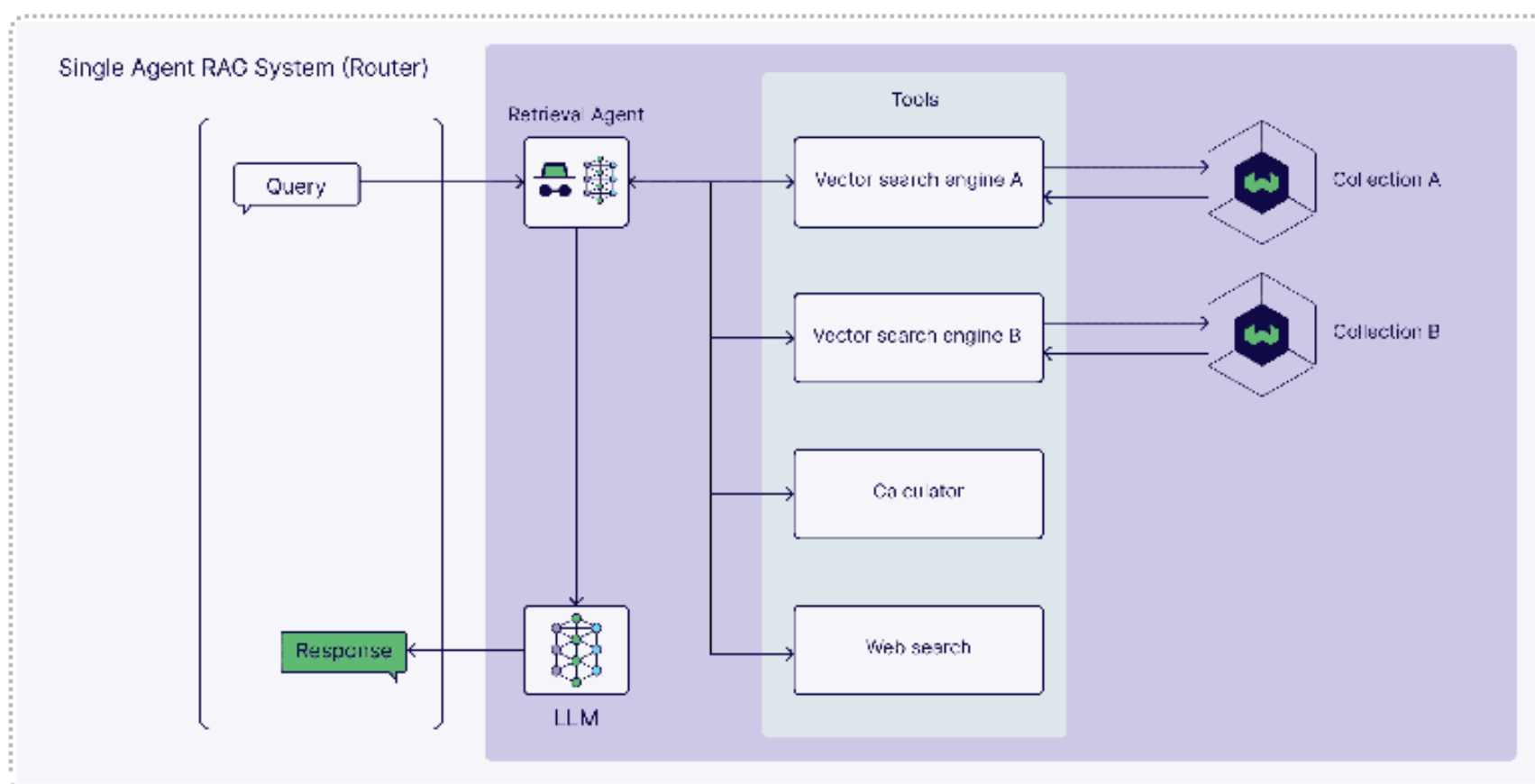
Agentic Retrieval-Augmented Generation (RAG) is revolutionizing how we access information. It combines contextual retrieval with autonomous decision-making to enhance user interactions.

What we'll cover?

- **Contextual Retrieval:** Agents intelligently determine when to access databases.
- **Autonomous Pipelines:** Agents optimize data retrieval processes without human intervention.
- **Enhanced Decision-Making:** Improves response relevance and efficiency.
- **Advanced Forms:** Includes Router RAG, Corrective RAG, and RAG Adaptive for specialized applications.



What is Agentic RAG?



Agentic RAG is an advanced form of Retrieval-Augmented Generation where agents autonomously decide when and how to query external data sources. This enhances traditional retrieval methods by optimizing the information retrieval process.

How it works

- Agents evaluate context to determine retrieval necessity.
- They autonomously query relevant databases.
- Responses are generated based on retrieved content and user inputs.

Key example

A customer service agent uses Agentic RAG to fetch relevant information based on user inquiries, improving response accuracy.



Characteristics of Agentic RAG

Autonomy in Querying

- Agents independently decide on data retrieval.
- This leads to more efficient information access.
- **Example:** A chatbot autonomously fetches data based on user context.

Contextual Understanding

- Agents assess the context of queries.
- They optimize retrieval based on user needs.
- **Example:** A support agent tailors responses based on previous interactions.

Dynamic Adjustment

- Agents adapt retrieval strategies over time.
- Continuous learning enhances response accuracy.
- **Example:** A recommendation system evolves based on user behavior.



Applications of Agentic RAG

Use Case 1

Customer Support Automation

- **Context:** High volume of customer inquiries.
- **Application:** Chatbots use Agentic RAG to provide accurate responses.
- **Benefits:** Reduced response time and improved customer satisfaction.

Use Case 2

Multi-Source Integration

- **Context:** Diverse data sources available.
- **Application:** Research tools combining various studies.
- **Benefits:** Comprehensive insights from multiple perspectives.

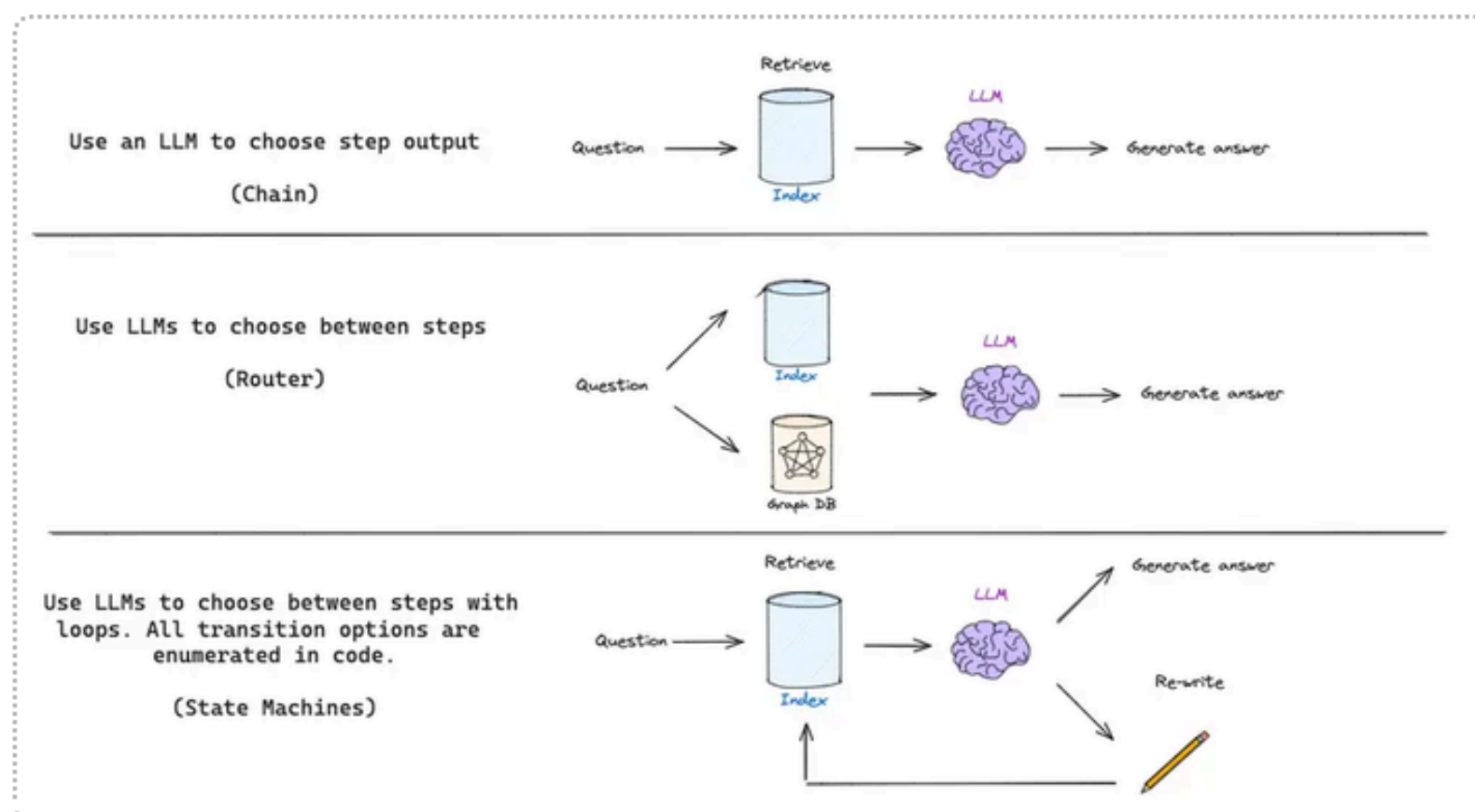
Use Case 3

Data Analysis

- **Context:** Large datasets requiring insights.
- **Application:** Agents autonomously query databases for analysis.
- **Benefits:** Faster decision-making and actionable insights.



Types of Agentic RAG

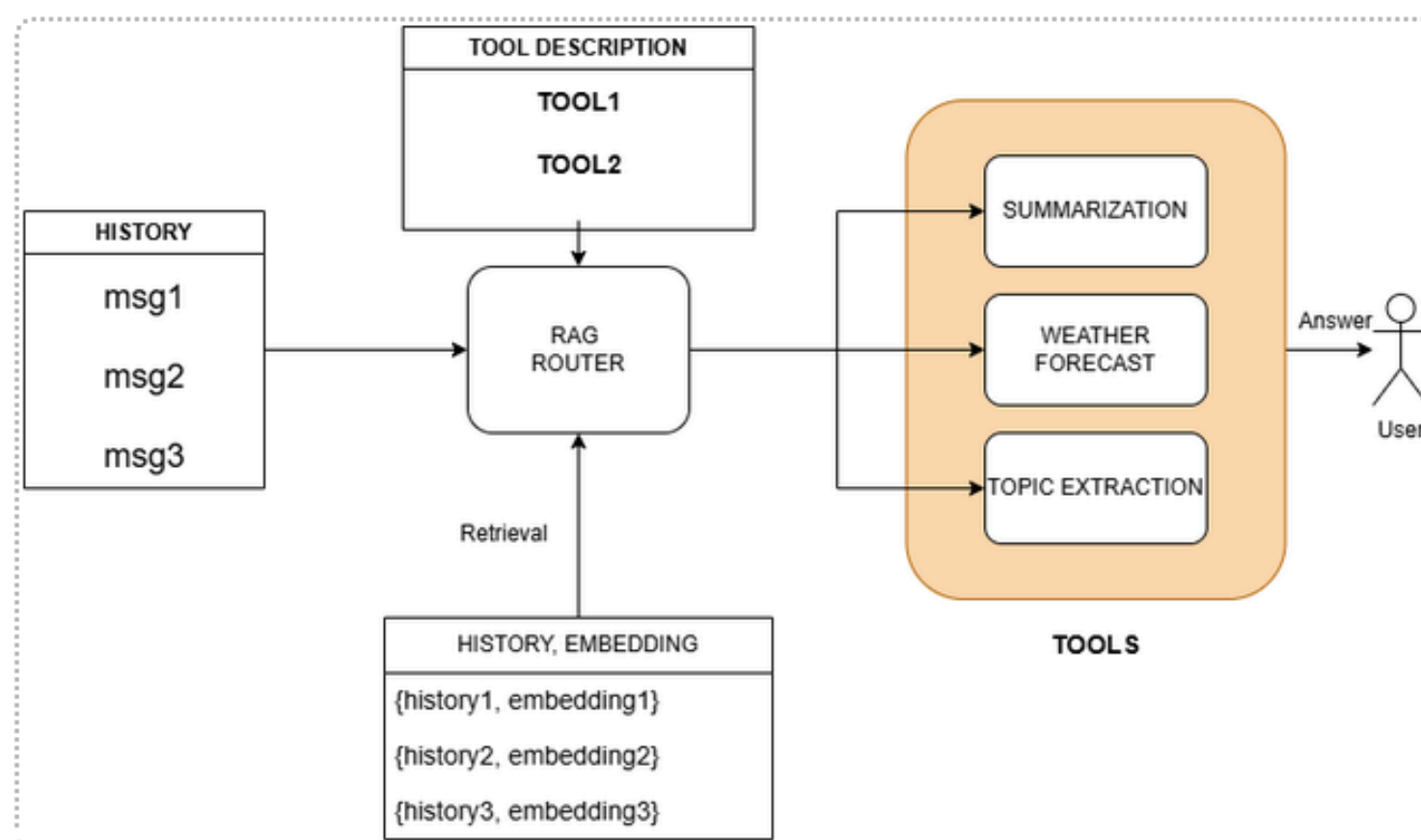


Agentic RAG encompasses various specialized types to enhance retrieval processes.

- **Router RAG:** Routes queries to appropriate data sources based on context.
- **Corrective RAG:** Refines retrieved information for accuracy before response generation.
- **Adaptive RAG:** Adapts retrieval strategies based on user and interactions feedback.
- **Reflective RAG:** Continuously evaluates its own outputs to future improve retrieval and reasoning.
- **Collaborative RAG:** Integrates multiple agents that coordinate to provide richer and more diverse insights.



Characteristics of Router RAG



Router RAG intelligently directs queries to the most relevant data sources based on context.

Key characteristics

- **Dynamic Routing:** Adapts query paths in real-time.
- **Context-Aware:** Considers user context for optimal data access.
- **Multi-Source Querying:** Integrates information from various databases.

Best for

Ideal for scenarios with multiple data sources needing context consideration.



Use Cases for Router RAG

Use Case 1

Real-Time Data Retrieval

- **Context:** Immediate access to updated information.
- **Application:** News aggregators fetching the latest articles.
- **Benefits:** Timely updates for users.

Use Case 2

Multi-Source Integration

- **Context:** Diverse data sources available.
- **Application:** Research tools combining various studies.
- **Benefits:** Comprehensive insights from multiple perspectives.

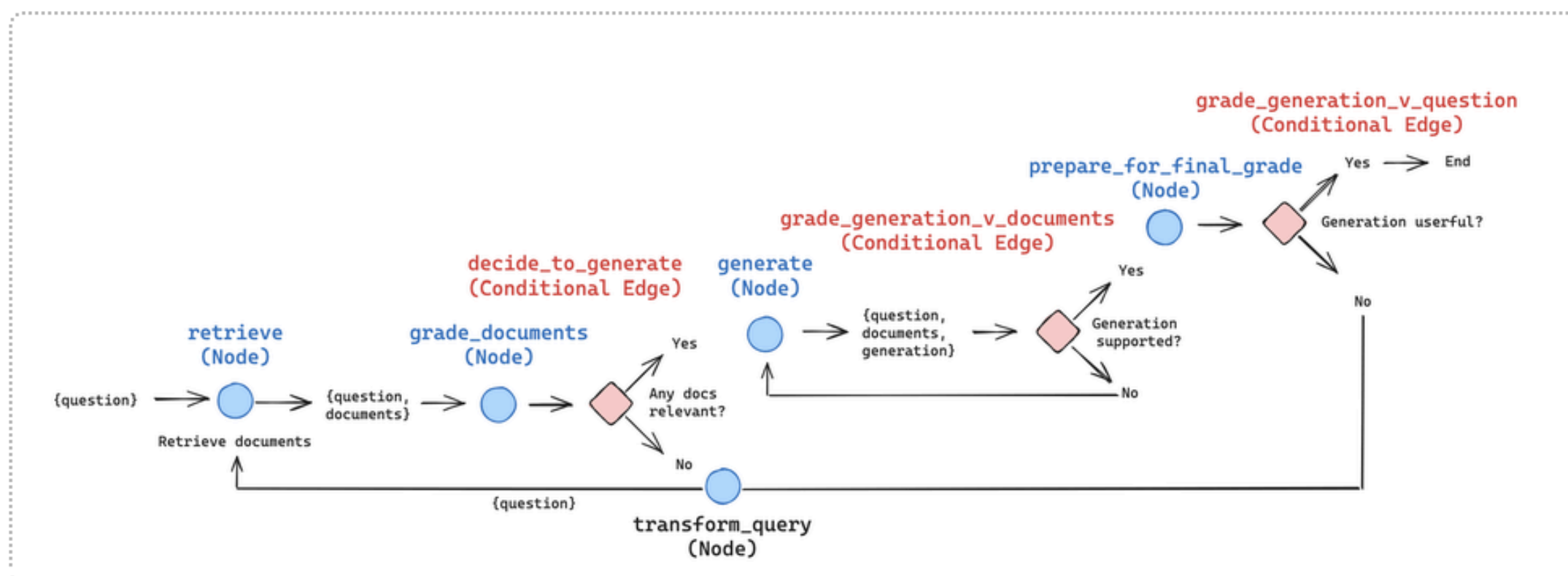
Use Case 3

Dynamic Content Delivery

- **Context:** Personalized user experiences.
- **Application:** E-commerce platforms tailoring product recommendations.
- **Benefits:** Increased user engagement and sales.



Understanding Self-RAG



Self-RAG is an advanced form of RAG where agents autonomously evaluate and adjust their retrieval strategies.

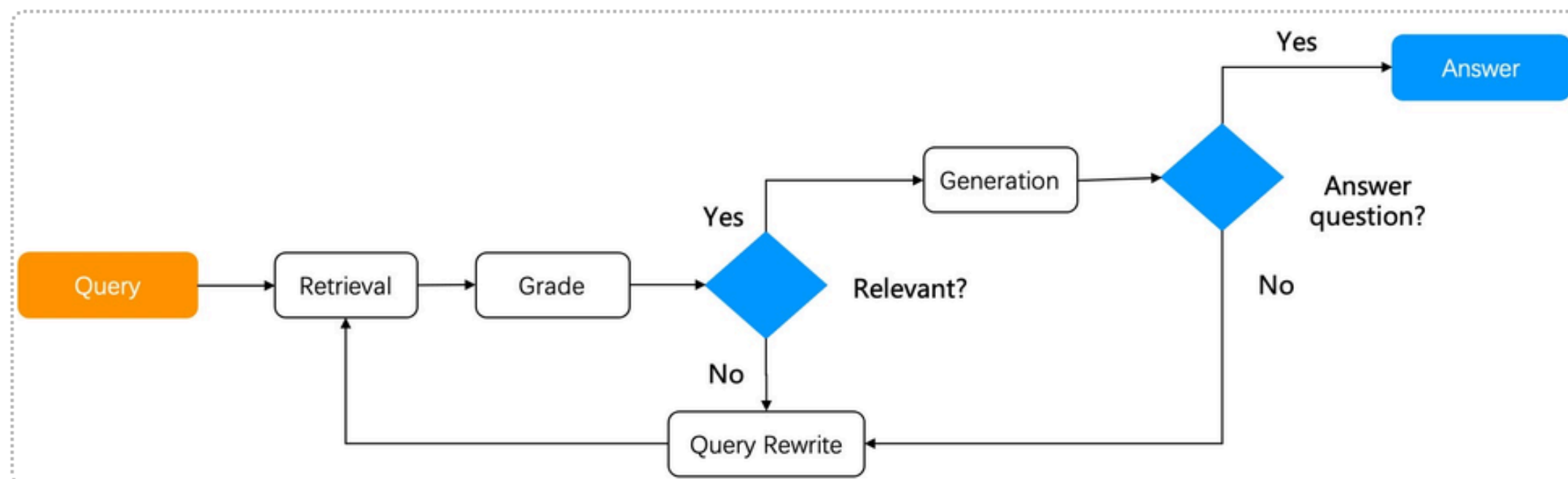
Key characteristics

Self-RAG is crucial for applications requiring ongoing improvement in information retrieval.

- **Self-Evaluation:** Agents assess their performance.
- **Autonomous Adjustments:** They modify strategies based on outcomes.
- **Feedback Loops:** Continuous learning from interactions enhances future retrieval.



Real-World Example of Self-RAG



Chatbots are increasingly used for customer interactions.

The Challenge

Ensuring the chatbot provides relevant and personalized responses.

The Solution

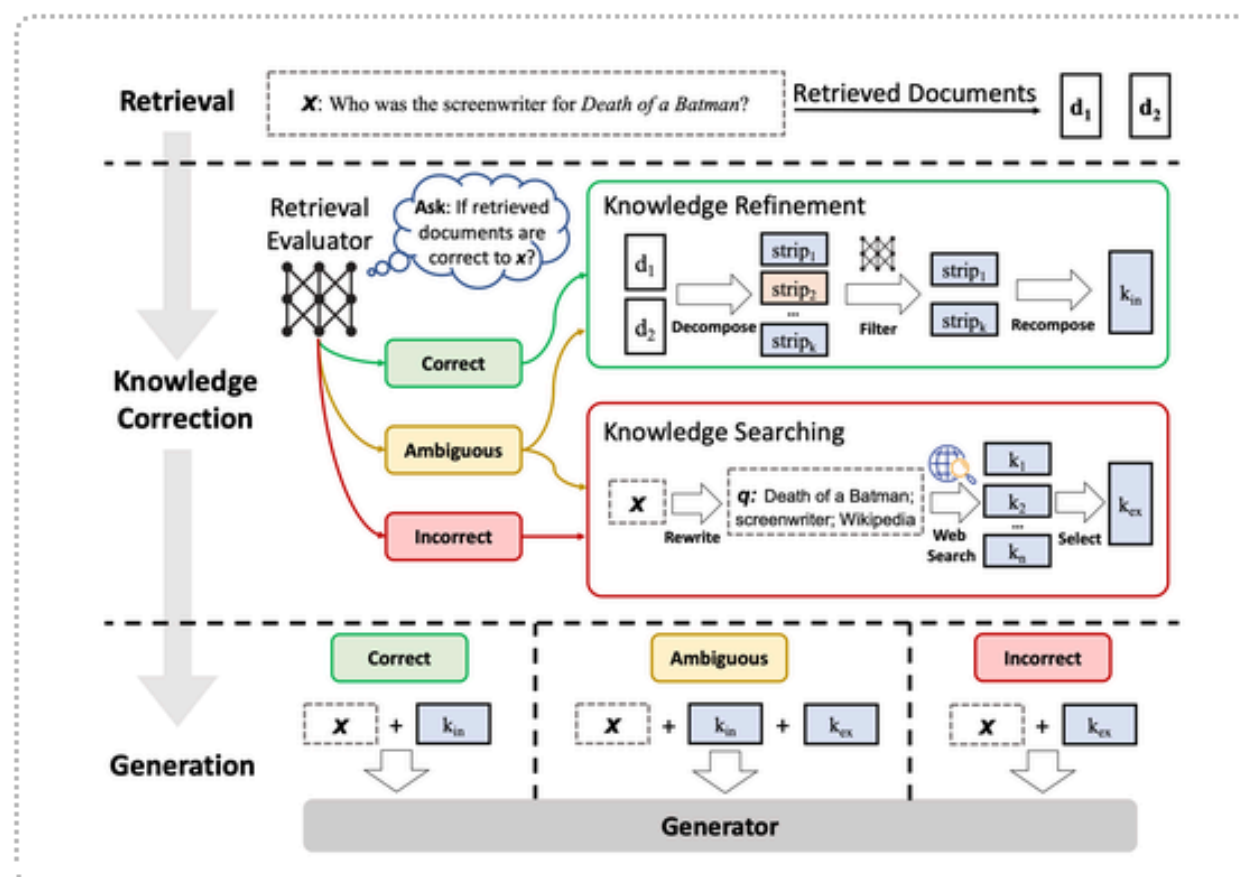
- The chatbot learns from previous interactions.
- It adjusts its retrieval strategies based on user feedback.
- Continuous updates improve response accuracy.
- Users receive tailored content recommendations.

Outcome

The chatbot's effectiveness increases, leading to higher user satisfaction and engagement.



Comparing RAG Types



Router RAG

Dynamic routing based on context.

Multi-source querying capabilities.

Best for: Scenarios with diverse data sources needing context consideration.

Corrective RAG

Error correction through feedback loops.

Increases accuracy of retrieved information.

Best for: Applications where accuracy is critical, like medical fields.



Graph-RAG: Structuring Knowledge

Graph-RAG structures knowledge into graph-based retrieval pipelines, enhancing the efficiency of information access.

Key characteristics

- **Graph-Based Structuring:** Organizes information in nodes. interconnected
- **Semantic Relationships:** Understands context through relationships.
- **Enhanced Query Capabilities:** Improves retrieval accuracy based on user intent.

Best for

Ideal for applications where understanding relationships between data points is crucial for accurate retrieval.